



BACKLOG SUBJECT EXAMINATIONS - FEBRUARY 2024

B.E. - Artificial intelligence and Data		
Program	: Science / Artificial Intelligence and Machine Learning	Semester : IV
Course Name	: Design and Analysis of Algorithms	Max. Marks : 100
Course Code	: AD43 / AI43	Duration : 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) Compute the stable matching set S for the following set of men and women using Gale-Shapley algorithm. CO1 (08)

Men's preference list

W	H	I	J	K
X	I	J	H	K
Y	J	H	I	K
Z	H	I	J	K

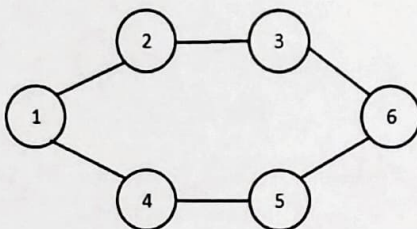
Women's preference list

H	X	Y	Z	W
I	Y	Z	W	X
J	Z	W	X	Y
K	W	X	Y	Z

- b) Explain any three asymptotic properties of algorithms with an example for each. CO1 (06)
- c) Describe an algorithm to compute the distance between two points in a plane and deduce the time complexity for the same. CO1 (06)
2. a) i. Compare between perfect matching and stable matching? Consider an execution of the G-S algorithm that returns a set of pairs S for the given set of men M and women W. Prove that "The set S is a stable matching". CO1 (08)
- ii. Prove the following.
- i) $T(n) = 4n^2 + 3n + 2$ is $O(n^3)$
- ii) $T(n) = 4n^2 + 3n + 2$ is $\wedge(n)$
- b) Define 'Big o, Omega and theta' asymptotic notations for algorithms with an example. CO1 (06)
- c) Write an algorithm for selection sort and show that its time complexity is quadratic. CO1 (06)

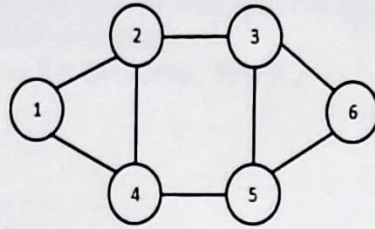
UNIT - II

3. a) Describe an algorithm to determine whether a given graph is bipartite or not. Using the same algorithm determine whether the given graph is bipartite or not. CO2 (08)



- b) Write the BFS traversal for the following graph.

CO2 (06)



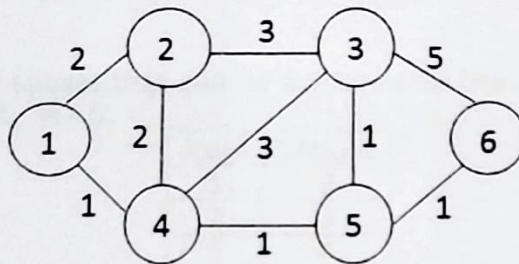
- c) Prove that the time complexity of merge sort is $O(n \log n)$ using "unrolling method".

CO2 (06)

4. a) Write an algorithm for topological ordering. Analyze its time efficiency. CO2 (08)
 b) Identify the number of inversions for the array $\{1,5,4,8,10,2,6,9,12,11\}$ using divide and conquer method. CO2 (06)
 c) Describe an algorithm to construct DFS traversal for a given graph $G=(V,E)$. CO2 (06)

UNIT - III

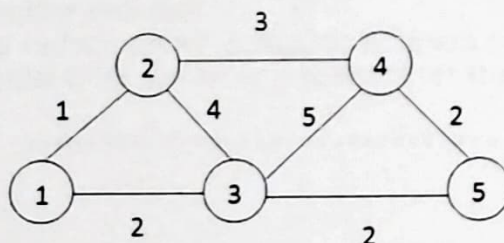
5. a) Find the shortest path from source node 1 to all other nodes for the following graph using Dijkstra's algorithm. CO3 (10)



- b) Identify the minimum spanning tree for a given graph $G=(V,E)$ using Prim's algorithm. CO3 (05)
 c) Define fixed-length encoding, variable-length encoding with an example. CO3 (05)
6. a) Identify the optimal set of intervals that can be scheduled for the following set on intervals using greedy interval scheduling algorithm. CO3 (08)

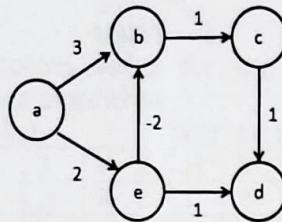
Intervals	Start time	Finish time
A	1	4
B	1	6
C	4	7
D	7	11
E	9	13
F	13	16

- b) Explain the concept of greedy technique for Prim's algorithm. CO3 (06)
 c) Identify minimum spanning tree for the following graph using Prim's algorithm with source node as 1. CO3 (06)



UNIT – IV

7. a) Identify the set of optimal set of intervals with maximum value for the given set of intervals and weights CO4 (06)
 Job Details {Start Time, Finish Time, Profit}
 Job 1: {10, 13, 5}
 Job 2: {13, 15, 2}
 Job 3: {16, 15, 10}
 Job 4: {15, 20, 20}
 Job 5: {16, 20, 10}
 Job 6: {21, 30, 10}
- b) Compare the approaches 'Dynamic programming' and 'Greedy' for problem solving. CO4 (06)
- c) Identify the shortest path from the source vertex a using Bellman ford algorithm. CO4 (08)



8. a) Find the optimal subset that can be achieved for the following items with maximum capacity $W=6$. CO4 (06)

Item	Weight
1	2
2	3
3	4

- b) Describe an iterative procedure to find the optimal weight for a given set of n intervals with their weights ' w '. CO4 (06)
- c) Identify the maximum value possible by considering the following instance and knapsack capacity $W=10$. CO4 (08)

Item	Weight	Value
1	4	40
2	7	42
3	5	20
4	3	12

UNIT- V

9. a) Describe Travelling salesman problem. Prove that travelling salesman problem is NP-Complete. CO5 (10)
- b) Compare Independent set and Vertex cover problems with an example. Are they both polynomial time reducible? Justify. CO5 (10)
10. a) Give an example of an NP complete problem. Explain the strategy used for solving NP complete problem. CO5 (10)
- b) Suppose you could reduce an NP complete problem to a polynomial time problem in polynomial time. Develop a solution for the consequence? CO5 (10)
